Deshik Reddy Putluru

Computational Mechanics | Multiphysics Modeling | FEM/CFD | Scientific Machine Learning

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I am a Ph.D. candidate developing high-fidelity finite-element models for multiphysics systems. My research uses numerical simulations to understand the biomechanics of cerebrospinal fluid flow and drug delivery in the spinal canal. Alongside these high-fidelity models, I build reduced-order models and am developing scientific machine learning surrogates to make patient-specific simulations practical for clinical and pharmaceutical decisions.

Education

Purdue University West Lafayette, IN, USA

Ph.D. in Mechanical Engineering | GPA: 3.89/4.00 *Jan. 2023 – Expected Dec. 2027*

Thesis: *Computational Modeling of Fluid–Structure Interaction and Drug Transport in the Spinal Canal*

Advisor: Prof. Hector Gomez

Committee: Prof. Adrian Buganza Tepole, Prof. Pavlos Vlachos, Prof. Vitaliy Rayz

Indian Institute of Technology Roorkee Roorkee, India

B.Tech. in Mechanical and Industrial Engineering, with Honors *Aug. 2018 – May 2022*

GPA: 8.98/10 (94.8%; Branch Rank: 2) | Thesis: *Shape Optimization of Savonius-Style Vertical-Axis Wind Turbines*

Advisor: Prof. Sushanta Dutta

Publications

Putluru, D.R., Tepole, A.B., & Gomez, H. Mixed-dimensional fluid–structure interaction simulations reveal key mechanisms of cerebrospinal fluid dynamics in the spinal canal. *Fluids and Barriers of the CNS* **22**, 81 (2025).

Srinivasan, A., **Putluru, D.R.**, Jones, E.A.V., & Gomez, H. Interstitial flow in the chick yolk sac exhibits organ-scale patterns driven by segregated leakage and drainage. *Under review*, *Proceedings of the National Academy of Sciences (PNAS)* (2026).

Putluru, D.R., Li, Z., & Gomez, H. Quantifying the role of tissue compliance in intrathecal drug transport using fluid–structure interaction simulations. *Submitted* (2026).

Putluru, D.R. & Gomez, H. Human in silico trial of intrathecal injection sites for AAV9 gene therapy in amyotrophic lateral sclerosis. *In preparation* (2026).

Research Experience

Computational Modeling of CSF Dynamics and Drug Transport West Lafayette, IN, USA

Graduate Research Assistant, Gomez Lab, Purdue University *May 2023 – Present*

Lilly Purdue Innovation Institute (LPII)

Fluid–Structure Interaction in the Spinal Canal *May 2023 – Jan. 2025*

- Developed a mixed-dimensional finite-element formulation that avoids fully volumetric FSI, enabling full-length patient-specific spinal simulations at approximately **60× lower cost**.
- Implemented residual-based VMS stabilization to model the effect of unresolved scales, linearized the coupled nonlinear equations, and built a **custom Newton–Krylov solver** using PETSc.
- Formulated a **deformation-Jacobian-based stiffening model** for ALE mesh motion that suppresses element inversion under large deformation and near-contact in narrow fluid gaps, enabling simulations through severe geometric distortion.
- Developed and benchmarked a **custom preconditioner** for the mixed-dimensional FSI problem, achieving a 3× speedup over PETSc's GAMG (algebraic multigrid) preconditioner.

Intrathecal Drug Transport and Injection-Protocol Optimization *Feb. 2025 – Present*

- Developed a **multicomponent finite-element solver** for intrathecal drug transport, with SUPG stabilization and a **custom discontinuity-capturing operator** for the advection-dominated problem.
- Adapted and implemented the **coupled-momentum method** for small-deformation spinal FSI to avoid ALE mesh motion, **reducing computational cost by approximately 30×** and enabling simulations over long time scales.
- Investigated injection-induced CSF pressure dynamics and quantified how drug solution baricity, gravity, and body posture affect intrathecal drug distribution.
- Developed a **reduced-order transport model** that retains the dominant CSF transport mechanisms, reducing compute time for a 1-hour drug-transport prediction from **20 hours to 1 second**.
- Enabled in silico evaluation of **AAV injection protocols for ALS**, providing drug-delivery insights to inform the design of in vivo nonhuman primate (NHP) trials.

Full-Field Surrogate of Spinal Drug Transport *Oct. 2025 – Sept. 2026*

- Guided the problem formulation, physical constraints, and input-parameter selection for a full-field surrogate (digital twin) of intrathecal drug transport; manuscript in preparation.
- Helped wrap the high-fidelity model for automated cluster runs and conduct 200 simulations with input parameters sampled from a Sobol sequence.
- Reduced the dimensionality of the simulation data with principal component analysis (PCA) for training the Gaussian-process surrogate.

Interstitial Flow in the Chick Yolk Sac West Lafayette, IN, USA

Research Collaborator, Gomez Lab, Purdue University *Aug. 2025 – Aug. 2026*

- Collaborative imaging and computational modeling study of coupled blood and interstitial flow across the whole yolk sac, with A. Srinivasan, E.A.V. Jones, and H. Gomez; contributed research ideas and the choice of methods. Manuscript under review at *PNAS*.

Physics-Constrained AI for Whole-Lung Digital Twin West Lafayette, IN, USA

Research Proposal Contributor, Purdue University *Jun. 2026 – Aug. 2026*

- Developed a **reduced-order (0D) whole-lung model of airflow, particle transport, and deposition** and used analytical methods and literature evidence to simplify deposition physics for biodistribution predictions.
- Proposed a **geometry-aware surrogate framework** using transferable foundation models to retain 3D flow details and enable **real-time prediction and design-space exploration** across airway geometries.
- Drafted the technical section of the research proposal.

Physics-Informed Neural Network Models in JAX West Lafayette, IN, USA

Course Project, Purdue University *Jan. 2025 – May 2025*

- Implemented label-free physics-informed neural network surrogates for fluid flows in JAX/Equinox (following Sun et al., 2020), embedding boundary conditions in the solution ansatz and using `jit` and `vmap` for compiled and vectorized residual evaluation.
- Computed Navier–Stokes residuals using JAX automatic differentiation and validated the surrogates against analytical Poiseuille flow and CFD solutions in **aneurysm and stenosis** geometries.
- Implemented an adaptive Newton-type optimizer for neural approximations of discontinuous advection solutions.

Modeling of Transient Film Boiling Using the Volume-of-Fluid Method Roorkee, India

Undergraduate Research Assistant, Indian Institute of Technology Roorkee *Jul. 2021 – Dec. 2021*

- Developed an OpenFOAM finite-volume solver using the volume-of-fluid method and phase-change physics to simulate transient film boiling around a heated metal sphere in a quiescent liquid pool.
- Analyzed vapor-film evolution and compared numerical predictions with available experimental measurements and analytical benchmarks.

Shock-Wave Propagation Through a Bifurcation Roorkee, India

Course Project, Indian Institute of Technology Roorkee *Aug. 2021 – Nov. 2021*

- Recreated a shock-tube bifurcation experiment in OpenFOAM and validated pressure and flow predictions against measurements; quantified how branch angle affects junction overpressure and flow structures to support safety analysis of blast-induced waves and hazardous gusts in tunnel networks.

Lamb-Wave Propagation Through Elbow Pipes Roorkee, India

Undergraduate Research Assistant, Indian Institute of Technology Roorkee *Feb. 2021 – May 2021*

- Developed a hybrid **semi-analytical finite-element (SAFE) framework** in MATLAB and COMSOL to characterize Lamb-wave propagation through elbow pipes, with applications to detecting crack-induced wave anomalies in large, difficult-to-inspect pipelines.

Professional Experience

Graduate Teaching Assistant, Purdue University *Jan. 2023 – May 2023*

- Assisted with a 180-student Micromechanics of Materials course: held office hours, guided students through assignments, and helped prepare problem sets, solutions, and instructional materials.

Business Analyst, ICICI Bank *Jul. 2022 – Oct. 2022*

- Evaluated production machine-learning models for auto-loan decisions, including performance and approval logic.
- Queried lending and model-output datasets in SQL to assess credit policies and decision thresholds.

Technical Skills

Modeling	Physics-constrained AI, multiphysics and multiscale modeling, fluid-structure interaction, interface dynamics, cavitation, phase-field modeling, scalar transport.
Numerical Methods	Computational mechanics, computational fluid dynamics, finite element/difference methods, variational multiscale methods, discontinuity capturing techniques, CutFEM.
Programming & Tools	C++, Python, MATLAB & Simulink, Linux, Slurm, Git, high-performance computing frameworks (MPI), PETSc, OpenFOAM, FEniCS, ANSYS (Structural, Thermal, Fluent), ABAQUS, SolidWorks, OpenCascade, ParaView, PyVista.

Presentations

Putluru, D.R. & Gomez, H. Computational modeling of fluid–structure interaction and drug delivery in spinal canal reveals key transport mechanisms. *Society of Engineering Science (SES) Annual Technical Meeting*, Oct. 2026. **Oral presentation**.

Poster presentation, Purdue One Health & Well-Being Research Event, March 2026.

Honors & Awards

- Professional Development & Innovation Award for Bachelor Thesis Project, presented by IIT Roorkee alumni, 2022.
- Ranked in the 99.64th and 97.55th percentiles in Joint Entrance Examination Main and Advanced, respectively (2018).
- Top 1% nationally, National Standard Examination in Junior Science (NSEJS), Indian Association of Physics Teachers (IAPT), 2015.

Leadership

American Society of Mechanical Engineers – Technical Head *ASME IIT Roorkee Chapter*

- Designed the mainframe and rear fork and led the end-to-end design and fabrication of the aerodynamic fairing.
- Optimized the fairing design for less drag using computational modeling, improving top speed by **28%**; team ranked **1st in design and 4th overall** in the ASME Human-Powered Vehicle competition.

SAE BAJA – Braking & Vehicle Dynamics Lead *Team KNOx, IIT Roorkee*

- Performed transient thermal and structural FEA of brake discs in ANSYS; optimized component geometry, designed an inboard caliper mount using topology optimization, and assessed resonance risk through modal analysis.
- Developed a full-vehicle braking model in MATLAB/Simulink using Simscape to predict single-stop braking performance and stopping distance across operating conditions.

Service & Outreach

- Student volunteer, Society of Engineering Science (SES) Annual Technical Meeting, Oct. 2026.
- Panelist, Eli Lilly–Purdue University Research Alliance Center (LPRC) Meet and Greet; represented the Gomez Lab and discussed research opportunities with students, Nov. 2025.
- Instructor, ASME IIT Roorkee workshop for 150+ students on CAD (SolidWorks) and FEA (ANSYS), Oct. 2020.